FemboyEngine2D — Справочник API

Движок на C++26 в парадигме data-oriented: полная публичная поверхность, контракты и пошаговый пример

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1. Область действия, соответствие стандарту и правила движка

Этот справочник описывает публичную поверхность FemboyEngine2D: все заголовки из Source/Engine и инфраструктуру редактора, с которой линкуется хост. Текст написан по дереву Release, поэтому описанный символ существует в поставляемых заголовках, а индекс в приложении генерируется из этих заголовков при каждом запуске.

Движок рассчитан на C++26 с GCC 16 или новее и Clang 19 или новее; релиз-кандидат под Windows собирается clang-cl со стандартной библиотекой MSVC. Четыре правила являются несущими, и каждый публичный API спроектирован вокруг них:

- Состояние симуляции — это простые trivially copyable данные. Ни виртуальных вызовов, ни владеющих указателей, ни std::string в компонентах, снапшотах и сетевых пакетах.
- Горячие пути не выделяют память. Кадровая память берётся из FrameArena, логи пишутся в фиксированные char-буферы, а колбэки — это указатель на функцию плюс void*, никогда std::function.
- Детерминизм — свойство данных, а не планировщика. Каждая подсистема отдаёт образ с контрольной суммой, и NetRollbackSession может воспроизвести только то, что воспроизводится байт в байт.
- Никакого RTTI в коде движка: dynamic_cast и typeid запрещены стилевым гейтом. Cast<> вместо этого обходит стабильные идентификаторы типов (см. раздел 3).

> Стилевой гейт (Source/Tools/Dev/CheckStyle.py) проверяет правила именования и структуры: PascalCase имена файлов, Allman-скобки, обязательные фигурные скобки у управляющих конструкций, префикс In у параметров, префикс Scoped у локальных переменных, отсутствие комментариев в конце строки и кириллицы в исходниках движка.

2. Сборка, цели и флаги компилятора

Одно дерево CMake производит три вида артефактов: статическую библиотеку движка (FemboyEngine2D::Engine), библиотеку и приложение редактора (FemboyEngine2D::Editor) и офлайн-инструменты. Поставляемая игра линкует только движок; Dear ImGui, ImGuizmo, ImPlot и TOML-писатель не попадают в поставочные цели.

Опция CMake	По умолчанию	Действие
FBY_BUILD_EDITOR	ON	Собирает fby_editor, приложение редактора и плагины панелей.
FBY_BUILD_TESTS	OFF	Собирает набор ctest в Source/Tests.
FBY_BUILD_TOOLS	ON	FempakCooker, LocTool, SceneTool, ShaderCompiler, AotCompiler.
FBY_BUILD_PLATFORM_PROVIDERS	ON	Плагины провайдеров Steam и Epic Games Store.
FBY_RHI_BACKEND	auto	auto, vulkan, metal или null. auto выбирается по платформе.

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo \
      -DFBY_BUILD_TESTS=ON -DFBY_BUILD_EDITOR=ON
cmake --build build --target fby_editor
ctest --test-dir build --output-on-failure
python Source/Tools/Dev/CheckStyle.py
```

Обёртка компилятора (cmake/FbyCompiler.cmake) задаёт флаг C++26, набор предупреждений и переключатели санитайзеров. Не добавляйте флаги отдельным целям: цель, которой нужен другой диалект, не должна существовать.

3. Базовые типы, FORCEINLINE и семейство приведений

Definitions.hpp — контрактный заголовок движка. Он содержит макросы экспорта, макросы утверждений (FBY_ASSERT, FBY_VERIFY, FBY_UNREACHABLE), механизм стабильных TypeId, семейство приведений и макросы регистрации компонентов. Остальные заголовки включают его прямо или транзитивно.

```
#if defined(_MSC_VER)
#define FORCEINLINE __forceinline
#elif defined(__GNUC__) || defined(__clang__)
#define FORCEINLINE inline __attribute__((always_inline))
#else
#define FORCEINLINE inline
#endif
```

FORCEINLINE — единственное написание инлайна в публичных заголовках. Он применяется к аксессорам, горячей арифметике и шаблонным приведениям и никогда не заменяет измеренную оптимизацию. Существуют три семейства:

Символ	Контракт
FemCast<To>(value)	Непроверяемый static_cast со сворачиванием ссылок. Используется, когда конкретный тип доказан конструкцией (выход фабрики, тег хранилища). Нулевая цена в рантайме.
Cast<To>(object)	Проверяемое приведение. Апкаст — compile-time static_cast; даункаст сравнивает GetTypeId() объекта с StableTypeId<To>() и цепочкой родителей; несвязанные типы не компилируются. При несовпадении возвращается nullptr, исключений и RTTI нет.
CastComponent<To>(registry, entity)	ECS-написание Cast<>: EnTT проверяет тип компонента на этапе компиляции, а реестр проверяет сущность. Возвращает nullptr, если компонента у сущности нет.

```
// Downcast through the stable type table, not through RTTI.
Fby::TypeObject* base = factory.Create("Sprite");
if (Fby::Sprite* sprite = Fby::Cast<Fby::Sprite>(base))
{
    sprite->Animate(dt);
}

// Registration binds Sprite to TypeObject and derives its stable id.
FBY_REGISTER_TYPE(Fby::Sprite, Fby::TypeObject);
```

StableTypeId<T>() — consteval FNV-1a от зарегистрированного имени типа (или сигнатуры компилятора, если имя не закреплено). Смена зарегистрированного имени меняет id, поэтому имена компонентов входят в сериализованную схему, а Snapshot v4 хранит LayoutHash рядом с именем.

4. FrameArena: единственный кадровый аллокатор

FrameArena — это bump-указатель по памяти, которой владеет вызывающий код. Он сбрасывается в начале кадра и никогда не освобождает отдельные выделения.

Системы, которым нужны временные массивы, берут его из SystemContext::frameArena; система, вызывающая operator new в кадровом пути, — дефект, а не стилистическое предпочтение.

```
Fby::FrameArena arena(storage);          // caller owns storage
arena.Reset();                           // once per frame
auto* items = arena.AllocateArray<MeshRenderItem>(count);
if (items == nullptr)
{
    // Arena exhausted: degrade, never allocate on the heap.
    return;
}
const std::size_t used = arena.Used();
const std::size_t capacity = arena.Capacity();
```

- Allocate() возвращает nullptr вместо исключения при исчерпании арены; вызывающий код обязан обработать отказ.
- AllocateArray<T>() проверяет std::is_trivially_copyable_v<T>: арена хранит только POD.
- Для типов с повышенным выравниванием нужно явно передать выравнивание; по умолчанию используется alignof(std::max_align_t).

5. JobSystem и потоковая диагностика

JobSystem владеет пулом std::jthread. Задача — это указатель на функцию, указатель на контекст и полуоткрытый диапазон, поэтому отправка задачи не выделяет память. ParallelFor делит диапазон на куски, позволяет каждому участнику брать следующий кусок из атомарного счётчика и всегда выполняет хотя бы один кусок на вызывающем потоке — так однопроцессорные машины остаются корректными, а повторный вход из воркера безопасным.

```
jobs.ParallelFor(entityCount, 4096, [&](std::uint32_t begin, std::uint32_t end)
{
    for (std::uint32_t i = begin; i < end; ++i)
    {
        Integrate(entities[i], dt);
    }
});
jobs.WaitIdle();

// Diagnostics: one row per thread, slot 0 is the main thread.
std::array<Fby::JobSystem::ThreadLoad, Fby::JobSystem::kMaxThreadSlots> loads{};
const std::uint32_t slots = jobs.SampleThreadLoad(loads);
```

ThreadLoad содержит busyNanoseconds (время внутри работы движка), taskCount и 16-байтное имя. Счётчики выровнены по кэш-линии на поток, поэтому два одновременно завершившихся воркера не конфликтуют. SceneDiagnosticsPanel превращает два замера в процент занятости каждого потока.

> ThreadSlotCount() равен ThreadCount() + 1 и ограничен фиксированным массивом MCounters; устройство с более чем 64 воркерами сообщает первые 64 слота.

6. ECS, планирование и снапшоты

Реестр — это EnTT. Системы регистрируются в SystemGraph в одной из четырёх фаз (PreUpdate, FixedUpdate, PostUpdate, Render). SystemContext — единственный параметр системы; он содержит реестр и JobSystem напрямую, а указатели на подсистемы сгруппированы в доменные бандлы (render, assets, audio, physics, text, accessibility, gameplay), поэтому система читает только объявленные бандлы. Запланированная система получает узкий view: `PhysicsSystemView`, `RenderSystemView`, `GameplaySystemView`, `AudioSystemView`, `UiSystemView`, `AccessibilitySystemView` или `CoreSystemView`, адаптированный через `Bind*System<&Function>()`.

```
Fby::Ecs::SystemGraph graph;
graph.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem, &physics);
graph.Add(Fby::Ecs::Phase::FixedUpdate, "movement", &MovementSystem, nullptr, 1ull <<
0);
graph.Add(Fby::Ecs::Phase::PostUpdate, "platform", &GogAchievementSystem, &gog);

Fby::Ecs::SystemContext context{registry, jobs};
context.physics = &physics;
context.window = &window;
graph.Build();
graph.Run(Fby::Ecs::Phase::FixedUpdate, context);
```

Snapshot v4 — канонический образ реестра с контрольной суммой. Компоненты с рантайм-хэндлами (b2BodyId, b2ShapeId) пишутся рукописными кодеками; снапшот никогда не сериализует эти хэндлы, а восстановление пересоздаёт тела из полей компонента. Схема хранит Mirror-имя и LayoutHash каждого компонента, поэтому изменение поля отклоняется, а не читается молча неверно.

API	Контракт
GetOrAdd<T>(registry, entity)	Добавляет компонент при отсутствии и возвращает стабильную ссылку.
ecs::SnapshotWriter / SnapshotReader	Пишет и валидирует канонический образ; контрольная сумма — FNV-1a по payload.
Fby::Net::NetRollbackSession	Использует snapshot v4 и воспроизводит тики при опоздании ввода (раздел 10).

7. RHI, элементы отрисовки и статистика кадра

RHI — рукописный слой Vulkan/Metal. Кадры записываются в постоянные кольцевые буферы; спрайты — это 48-байтные POD; меши рисуются одним indirect-вызовом на материал/слой; linear-blend skinning — один compute-диспатч. FrameStats показывает, что реально отправил проход, и именно это рисует SceneDiagnosticsPanel.

Поле FrameStats	Значение
items / batches / drawCalls	Спрайт-элементы, батчи и отправленные вызовы.
meshItems / meshIndirectCalls	Количество MeshRenderItem и indirect-вызовов.
skinDispatches / skinRanges	Диспатчи и диапазоны compute-скиннинга.
uploadedBytes	Байты, загруженные в этом кадре.

> Viewport редактора рендерит в offscreen-цель, а не в swapchain. EnsureViewportTarget() задаёт размер цели в физических пикселях; цель, заданная в точках, — обычная причина размытого Viewport на HiDPI-мониторе.

8. Материалы: кук графов, кэш пайплайнов и продакшн-путь

Авторинг материала и его компиляция находятся по разные стороны границы движка. `Render::MaterialGraph` — авторский DAG с фиксированными ёмкостями (128 узлов, 384 пина, 64 параметра), trivially copyable, хешируется без позиций в редакторе. `Render::MaterialGraphCompiler` превращает прошедший валидацию граф в текст GLSL 450. `fby_matc` запускает этот компилятор и glslang на этапе сборки и пишет payload `.femmat`, поэтому поставленная игра не линкует glslang и не разбирает GLSL. Полное обоснование — в [ADR 0002](../../adr/0002-materials-in-engine.md).

Компонент	Заголовок	Контракт
`Render::MaterialGraph`	`Engine/RHI/MaterialGraph.hpp`	Авторский DAG; `Hash()` — идентификатор пайплайна, позиции в UI не входят
`MaterialGraphCompiler`	`Engine/RHI/MaterialGraphCompiler.hpp`	DAG в GLSL 450 и флаги контракта шейдера
`EncodeMaterialGraph` / `DecodeMaterialGraph`	`Engine/RHI/MaterialAsset.hpp`	Wire-формат `.fgraph`: явные little-endian поля, без memcpy структуры
`EncodeCookedMaterial` / `DecodeCookedMaterial`	`Engine/RHI/MaterialAsset.hpp`	Payload `.femmat`: 32-байтный заголовок, SPIR-V вершинного и фрагментного этапов, декодирование без копирования
`Render::MaterialPipelineCache`	`Engine/RHI/MaterialPipelineCache.hpp`	128 пайплайнов по хешу графа, с подсчётом ссылок, LRU-эвикцией и без обращений к куче
`Rhi::RHI::SupportsRuntimeShaderPipelines`	`Engine/RHI/RHI.hpp`	Запрос возможности: Vulkan возвращает true, Null и Metal — false

```
# Author the graph once, cook it in the build step
build/bin/fby_matc assets/materials/tinted.fgraph -o assets/materials/tinted.femmat

// `InBytes` is the .femmat payload from the VFS or a package entry.
Render::CookedMaterialView ScopedMaterial;
if (!Render::DecodeCookedMaterial(InBytes, &ScopedMaterial))
{
    return false;
}

// One cache per device: acquire at load time, release at shutdown.
const Rhi::ShaderPipelineHandle ScopedPipeline = InMaterials.Acquire(InRhi,
ScopedMaterial);

Rhi::MeshRenderItem ScopedItem;
ScopedItem.materialPipeline =
```

```
ScopedPipeline.IsValid() ? ScopedPipeline.id : Rhi::kDefaultMaterialPipeline;
```

- Хеш графа — идентификатор пайплайна. Правка графа даёт новый хеш, поэтому старый пайплайн становится записью кэша без ссылок, а не меняется на месте.
- Маска контракта едет вместе с payload: сгенерированный вершинный этап, view projection во фрагменте, push-константы материала, varying world/vertex position. Живая запись, чей хеш приходит с другим контрактом, отклоняется: это другой шейдер.
- `Acquire()` добавляет ссылку, `Release()` снимает. Пайплайн с нулём ссылок остаётся резидентным, поэтому материал, берущийся раз в кадр, не пересоздаётся; он уничтожается только при LRU-эвикции, `ReleaseAll()` или `Shutdown()`.
- Кэш не аллоцирует и не форматирует текст на кадровом пути; `LastError()` возвращает фиксированный буфер ошибки.
- Бэкенды без рантайм-пайплайнов остаются на шейдерах движка. Спрашивайте `SupportsRuntimeShaderPipelines()`, а не определяйте это по невалидному хендлу.

Payload	Поля	Проверяется при загрузке
`.fgraph`	magic `FBYG`, версия, счётчики узлов/пинов/параметров, узлы, пины, параметры	magic, версия, счётчики против ёмкости, точный размер, виды узлов, индексы пинов и параметров
`.femmat`	magic `FBYM`, версия, маска контракта, хеш графа, длины вершинного и фрагментного этапов	magic, версия, размер против длин, оба SPIR-V модуля, согласованность контракта

Редактор сохраняет живое редактирование поверх того же кэша: `Fby::Editor::MaterialGraphRuntime` перекомпилирует граф при смене хеша, отдаёт ошибки по вкладкам и передаёт владение пайплайном в `MaterialPipelineCache`. `MaterialGraphRuntime::SetDomain` переключает живую компиляцию между mesh- и sprite-контрактом, а `SaveGraph`/`LoadGraph` панели пишут и читают payload `.fgraph`, который потребляет cooker. Поставленная сборка обходится без glslang и вызывает только `DecodeCookedMaterial` и `Acquire`.

Спрайты получают материалы через третий артефакт: `MaterialCompileOptions::domain` выбирает спрайтовый домен, где сгенерированный фрагментный этап совпадает с varying-ами `quad.vert` (`vUv`, `vColor`, `vLayer` и `vWorld`, если узел его требует), сэмплирует слой атласа элемента через `vLayer`. `Time` доступен: спрайтовый push-блок несёт часы материала по смещению 80, поэтому сгенерированный этап читает `pc.materialParams.x` так же, как меш-шейдер. `fby_matc --sprite` кукает такой вариант. `DrawOrder::MaterialBucket` помещает спрайт в бакет, `ChangeMaskRenderExtract::SetMaterialPipelines` сопоставляет бакеты с id кэшированных пайплайнов, а экстракт выдаёт один `DrawBatch` на бакет с уже разрешённым `DrawBatch::pipeline`. 64-битный ключ несёт бакет (16 слой, 8 материал, 32 глубина, 8 последовательность), поэтому материальный батч не

разрывается глубиной. Узлы, которым нужна меш-проекция (`Fresnel`, `VertexPosition`), отклоняются в спрайтовом домене с понятной причиной.

9. Физика: Box2D v3, суставы, ткань и SdfCollisionSystem

PhysicsSystem владеет одним b2WorldId и отдаёт его расширениям как упакованный uint32, поэтому box2d.h не попадает в публичные заголовки. Тела зеркалируются из компонентов RigidBody2D/Collider2D в Box2D, шагают с фиксированной частотой и читаются обратно перед интерполяцией. PhysicsJointSystem строит фиксированный граф соединений: цепь, верёвка и пружины с пределами прочности, накоплением повреждений, разрезами из геймплея, ремонтом, пересборкой на месте и связями. ClothSystem интегрирует Verlet-сетку и проецирует её на те же коллайдеры Box2D.

Контракт SdfCollisionSystem:

Член	Контракт
SetField(const SdfField&)	Принимает невладеющий mdspan на row-major сетку. Отрицательные значения — твёрдое тело; память поля принадлежит рендереру/рельефу, а не системе.
Step(registry, physics, fixedDt)	Сэмплирует сетку в углах и центре тела, превращает градиент в нормаль, применяет импульс выталкивания с учётом массы, отражает внутрь направленную скорость и пишет исправленное состояние обратно. Возвращает число исправленных тел.
Update(SystemContext&)	ECS-обёртка вокруг Step(); вызов без физики ничего не делает.
Contacts()	Фиксированный массив SdfContact {entity, point, normal, penetration, impulse, sequence}.
SdfCollisionDesc	pushOut, restitution, maxImpulse, subSamples, dynamicOnly.

```
Fby::SdfCollisionDesc desc;
desc.pushOut = 0.65f;
desc.restitution = 0.05f;
desc.subSamples = 2;
sdf.Initialize(desc);
sdf.SetField(terrain.Field());
sdf.Register(graph);                    // Phase::FixedUpdate

// Viewport overlay: red vectors whose length is the penetration.
debug.Build(context, Fby::Editor::PhysicsDebugSdf, &sdf);
```

> SdfCollisionSystem не входит в SystemContext. Хост, которому нужна SDF-физика, регистрирует систему явно через Register(graph) и хранит экземпляр живым; отладочный проход Viewport принимает тот же указатель.

10. NetRollbackSession: детерминированный роллбэк

Сессия связывает snapshot v4, CommandLog, Pcg32 и геймплейные компоненты (теги, атрибуты, эффекты и скиллы) в один контур роллбэка с фиксированной ёмкостью.

После Initialize() аллокаций нет: одно кольцо снапшотов, одно кольцо ввода на игрока, фиксированная карта сущностей и фиксированные метаданные кадров. Колбэк симуляции — указатель на функцию плюс void*.

Метод	**Контракт**
Initialize(registry, schema, config, arena)	Настройка без аллокаций с ареной снапшотов, принадлежащей вызывающему.
Initialize(registry, schema, config)	Удобная перегрузка; выделяет одну арену (единственная аллокация сессии).
SetSimulation(fn, user) / SetCommandLog(log)	Привязывает чистую функцию симуляции и лог структурных команд.
Start(rng)	Фиксирует базовый тик и сохраняет RNG сессии. Обязателен до Step().
Step(localInput)	Симулирует ровно один тик с предсказанием отсутствующего удалённого ввода и фиксирует снапшот.
SubmitRemoteInput(player, frame)	Сохраняет опоздавший или текущий ввод; кадр для уже симулированного тика помечает ожидающий роллбэк.
ReplayPending(report)	Восстанавливает снапшот перед самым ранним исправленным тиком, восстанавливает RNG и курсор CommandLog, пересимулирует, фиксирует новые снапшоты и сообщает изменённые тики.
CompareChecksums(tick, snap, state, out)	Сравнивает удалённый отчёт с сохранённым кадром; две контрольные суммы отделяют побайтовый desync от desync игрового состояния.

```
Fby::Net::NetRollbackSession::Config cfg;
cfg.playerCount = 2;
cfg.localPlayer = 0;
cfg.rollbackTicks = 64;
session.Initialize(registry, schema, cfg, arena);
session.SetSimulation(&SimulateTick, &game);
session.Start(pcg);

while (running)
{
    session.SubmitRemoteInput(remote, received);
    session.Step(localInput);
    if (session.RollbackPending())
    {
        Fby::Net::RollbackReport report;
        session.ReplayPending(&report);
        if (report.changedTicks > 0)
        {
            LogBoy().Log(Fby::Editor::LogBoySeverity::Desync, "net",
                         "rollback {}..{} changed {}", report.fromTick, report.toTick,
                         report.changedTicks);
        }
    }
}
```

- NetInputFrame — 24 байта целочисленных осей (int16 с фиксированной точкой). Float запрещён: битовый шаблон float не гарантированно переживает другой компилятор.
- Result перечисляет все режимы отказа (NotInitialized, BufferFull, RollbackTooOld, ChecksumMismatch, ...); ResultName() — написание для логов.
- Сессия некопируемая и владеет ареной, если использована удобная перегрузка.

11. Аудио: микшер, DSP-граф, лимитер

AudioSystem владеет устройством miniaudio и фиксированным микшером. DspGraph маршрутизирует узлы в шины, поддерживает дакинг и реверберацию и принимает команды через lock-free кольцо, поэтому игровой поток не блокирует аудиоколбэк. Limiter.hpp — финальная стадия true-peak.

- DeafMode глушит граф на корне Master и превращает звуковые триггеры в строки субтитров; это политика аудиосистемы, а не пост-эффект.
- Ogg/Opus декодируется вне аудиопотока; микшер видит только приготовленные .fpcm буферы.

12. PinkUI и WidgetCanvasSystem

PinkUI — игровой интерфейс: канвас фиксированной ёмкости, mdspan-раскладка сетки, разрешение якорей, hit-testing и упаковка в один батч. WidgetCanvasSystem превращает WidgetCanvas в FrameOutput (элементы отрисовки) без аллокаций; глифы берутся из атласа .fbyfont, сматпленного через VFS, поэтому поставочный рантайм не содержит TTF-растеризатора. Глифы существуют только в атласе: встроенного 8x8-фолбэка нет, а текстовый кадр без смонтированного массива шрифтов отклоняется, а не рисуется неверным слоем.

Член	Контракт
WidgetCanvasSystem::MountFonts(RHI&)	Обязательный этап холодного старта в главном потоке. Собирает GPU-массив из набора .fbyfont в VFS; ошибка — это PANIC в лог плюс FBY_ASSERT_MSG, а не фолбэк.
BuildFrame(context, canvas, output)	Раскладывает и отсекает один канвас, возвращая BuildStatus и заполняя FrameOutput. Текстовый канвас без актуального смонтированного атласа возвращает FontsNotMounted и ничего не пишет. Без аллокаций; безопасно вызывать из Phase::Render.
Update(context, canvases, output)	Собирает все активные канвасы в один вывод; хост отправляет результат одним батчем. Нарушение шрифтового контракта останавливает проход и ставит fontContractViolated.
WidgetCanvas::SetDisplayScale	Плотность из CameraStackSystem::DisplayScale; размеры шрифтов округляются до целых физических пикселей.
OledShiftState	Целочисленный 2-пиксельный сдвиг для защиты OLED; ApplyOledShift() привязывает его к пулу канвасов.

WidgetCanvasSystem — ещё и поверхность доступности: BlindMode читает подписи виджетов для нарративного лога, DeafMode получает виджеты субтитров, а матрицы цветокоррекции и FlashGuard применяются к цветам канваса перед упаковкой.

13. Ядро доступности (BlindMode, DeafMode, FlashGuard)

Доступность — это политика кадра, а не оверлей. BlindMode пропускает игровой проход, indirect-вызовы и диспатч скиннинга и публикует ограниченный нарративный лог. DeafMode глушит шину Master и превращает звуковые триггеры в SubtitleLines. FlashGuard сжимает резкие изменения яркости. Цветокоррекция использует опубликованные матрицы Machado 2009, а SrgbRelativeLuminance()/ContrastRatio() реализуют WCAG 2.2 для проверки при авторинге.

Элемент	Контракт
NarrativeEvent	SPSC-кольцо на 128 слотов с текстовым payload 160 байт; поток симуляции пишет, поток речи читает.
ColorVisionMatrix(deficiency)	Возвращает матрицу 3x3 для протанопии, дейтеранопии или тританопии.
ApplyColorVision(color, matrix, strength)	Применяет матрицу в линейном свете к sRGB Color32.
FlashGuard::Evaluate(luminance, dt)	Возвращает усиление для текущего кадра; большое изменение сжимается до порога.
EditorSettings::ide*	Те же политики применяются к самому окну редактора: коррекция Machado, высокий контраст WCAG, FlashGuard, навигация с клавиатуры и LogBoy как приёмник нарратива.

14. Конвейер ассетов, VFS и локализация

Ассеты адресуются относительным путём внутри VFS. Архив .fempak монтируется с приоритетом; каталог-монитор может его перекрывать — так редактор пишет loose-файл до того, как кукер вернёт его в архив. Read() возвращает span на mmap, который валиден до размонтирования или горячей перезагрузки файла.

Формат	Владелец	Назначение
.fempak	FempakCooker	Неизменяемый поставочный архив с индексом и опциональным шифрованием.
.fbyfont	FontAtlas (PinkUI)	Испечённый атлас глифов: [header][glyph records][pixels]; без TTF-растеризатора в рантайме.
.femeta	AssetDatabase	Типизированные метаданные, настройки импорта и стабильный AssetId.
.fbin	LocTool	Скомпилированная таблица локализации; сам интерфейс редактора — только английский.

> Интерфейс редактора намеренно не локализуется: меню, заголовки панелей, ошибки и инспектор — только на английском. Локализация относится к игровому контенту и документации, где таблица .fbin и руководство EN/RU покрывают поставляемые языки.

15. Платформенные сервисы: Steam, Epic Games Store, GOG

PlatformService — мост на стороне движка; PlatformAPI.h — плоский чистый C ABI, который реализует плагин SDK магазина. Провайдер загружается динамически, резолвит только документированные точки входа и преобразует все типы магазина на границе, поэтому игра никогда не включает steam_api.h, EOSSDK или заголовки Galaxy.

Провайдер	Статус в этом дереве	Примечания
SteamProvider.cpp	Реализован	Плоский API Steamworks через таблицу функций.
EgsProvider.cpp	Реализован	Провайдер Epic Online Services через тот же C ABI.
GogProvider.cpp	Реализован (шим)	Загружает galaxy64.dll / libgalaxy64.so / libgalaxy64.dylib во время работы и резолвит плоский ABI шима fby_gog_*. Шим поставляет игра, собранная под свою версию SDK. Galaxy — библиотека C++-интерфейсов, поэтому шим обязателен. GogAchievementSystem и GogStorageSystem пишут бинарный PlatformCommandLog в Phase::PostUpdate.
fby_fontc	Реализован	Офлайн-кукер TTF -> .fbyfont (stb_truetype, RGBA8-атлас, проверяемый заголовок).

```
// Pure C ABI: the provider is a shared library, not a C++ dependency.
typedef struct FbyPlatformApi {
    uint32_t version;
    void* user;
    bool (*init)(void* user);
    void (*shutdown)(void* user);
    void (*unlockAchievement)(void* user, const char* id);
    bool (*writeCloudFile)(void* user, const char* name, const uint8_t* data, uint32_t
size);
} FbyPlatformApi;
```

16. Редактор: дашборд, шрифты, LogBoy, Scene Diagnostics

Дашборд — единственная поверхность редактора, существующая до открытия проекта. Он устанавливает набор шрифтов редактора до первого кадра, поэтому первое окно уже отрисовано Inter с учётом физического DPI. Новый проект создаётся в

Projects/ в корне движка, путь определяет DefaultProjectsRoot() — никогда рабочий каталог процесса и никогда каталог установки.

Компонент	Контракт
EditorFonts	Загружает Inter и JetBrains Mono из mmap VFS. Субпиксельные продвижения (PixelSnapH=false), авто-оверсэмплинг, без bitmap-fallback, динамический атлас печётся под плотность пикселей вьюпорта. Устаревшие бэкенды получают RasterizerDensity + обратный FontGlobalScale.
ProjectDashboard	Initialize(paths, config, visible, window, vfs, fonts, uiScale): устанавливает типографику, определяет расположение Projects/ по умолчанию и создаёт его.
LogBoySystem / LogBoyPanel	Фиксированное кольцо на 4096 записей с текстом 192 байта, уровнями Trace/Info/Warning/Critical/Desync, FNV-1a хэшами систем и фильтрацией по цвету и хэшу.
SceneDiagnosticsPanel	Столбцы и тепловая карта занятости потоков, байты FrameArena, количество MeshRenderItem, indirect-вызовы, draw calls и точное время симуляции в микросекундах с avg/p99/peak.
PhysicsDebugVisualizer	Геометрия линий и точек в фиксированных буферах для тел Box2D, верёвок, ткани и нормалей SDF, рисуется поверх Viewport через карту мир→экран, заданную вызывающим.

> Атлас .fbyfont — испечённый ассет PinkUI и не может питать векторный растеризатор ImGui. Поэтому редактор загружает TTF Inter из того же монтирования VFS; заголовок атласа всё равно валидируется при старте, чтобы испорченный атлас игрового UI обнаруживал редактор, а не игрок.

17. Пошагово: игра с пустой папки

Этот пример создаёт играбельный проект со спрайтом, физикой, вводом с клавиатуры, HUD на PinkUI и детерминированной rollback-сессией. Каждый шаг соответствует реальному заголовку из индекса.

- 1. Запустите редактор и создайте проект через New Project. Дашборд создаёт <engineRoot>/Projects/<name> с assets/{textures,audio,scenes}, configs/ и project.femproj.
- 2. Импортируйте спрайт в assets/textures. Наблюдатель импортирует файл, пишет .femeta и готовит слой текстурного массива .ftex.
- 3. Создайте assets/scenes/main.toml с одним SpriteRenderer, одним Transform2D и одним RigidBody2D + Collider2D.
- 4. Зарегистрируйте системы: ввод в PreUpdate, физика в FixedUpdate, движение в Update, PinkUI в Render.
- 5. Дайте камере CameraStackSystem и вызовите SetDisplayScale() из SDL_GetWindowDisplayScale, чтобы глифы UI были чёткими на HiDPI.
- 6. Подключите AccessibilityKernel из настроек проекта; BlindMode и DeafMode читаются из project.femproj, а не зашиты в код.

- 7. Создайте NetRollbackSession на матч, привяжите чистый колбэк SimulateTick и подавайте значения NetInputFrame.
- 8. Логируйте через LogBoy: уровень + хэш системы + сообщение в фиксированном формате. Откройте панель LogBoy и фильтруйте по Desync при тестировании сети.
- 9. Включите Physics Debug Draw во Viewport при настройке коллайдеров и ткани.
- 10. Соберите архив через FempakCooker, прогоните ctest и стилевой гейт, затем поставляйте цель только с движком.

```
// 4-7 in code: systems, accessibility and rollback for one match.
Fby::Ecs::SystemGraph graph;
graph.Add(Fby::Ecs::Phase::PreUpdate, "input", &InputSystem);
graph.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem, &physics);
graph.Add(Fby::Ecs::Phase::FixedUpdate, "movement", &MovementSystem, nullptr, 1ull <<
1);
graph.Add(Fby::Ecs::Phase::Render, "ui", &UiSystem, &ui);

Fby::AccessibilityKernel kernel;
kernel.Configure(settings);              // blind/deaf/colour/flash

Fby::Net::NetRollbackSession session;
if (session.Initialize(registry, schema, cfg, arena) !=
Fby::Net::NetRollbackSession::Result::Ok)
{
    LogBoy().Log(Fby::Editor::LogBoySeverity::Critical, "net", "rollback init failed");
}
```

18. Тесты, стилевой гейт и чек-лист релиза

Набор намеренно не использует фреймворк: каждый тест — исполняемый файл с main(), возвращающим ненулевой код при ошибке, поэтому одинаково работает на headless CI-агенте, консольном devkit и рабочей станции разработчика. На релиз-кандидате проходит весь набор.

```
ctest --test-dir build --output-on-failure          # the full suite
python Source/Tools/Dev/CheckStyle.py               # style gate
python Documentation/Tools/GenerateAPIReference.py  # this manual
python Documentation/Tools/GenerateDocumentation.py
python Documentation/Tools/GenerateApplicationGuide.py
```

- Релизная сборка включает утверждения и один раз прогоняет весь набор с FBY_ENABLE_ASSERTS=1, потому что на RC существовали две скрытые ошибки компиляции debug-сборки в JobSystem.
- POSIX-ветка WriteFileAtomically() и LogLine() должна компилироваться на Linux/macOS в CI; Windows-only сборка их не видела.
- Стилевой гейт должен сообщать ноль ошибок, прежде чем коммит может считаться релизным.

19. Открытая работа

Открытая работа отслеживается в Documentation/ROADMAP.md.

20. Приложение A - сгенерированный индекс символов

Генерируется из заголовков при каждом запуске. Символ из этого списка объявлен в поставляемом дереве; заголовок и строка — авторитетное расположение.

Editor/Infrastructure (115)

Символ	Вид	Заголовок	Строка
AIStateTreeCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	44
AnimationGraphCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	45
AnimationTimelinePanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	46
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	20
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	47
AssetImportService	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	30
AssetImportService	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	48
Callbacks	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	38
Change	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	102
Codec	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	220
Command	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	234
CommandKind	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	33
CommandStack	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	30
CommandStack	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	49
Config	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	27
Config	Infrastructure	Source/Editor/Infrastructure/	39

		AssetImportService.hpp	
Config	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	88
ContentBrowserPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	50
CookRequest	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	115
CookResult	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	126
EditorCamera	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	91
EditorDesc	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	62
EditorFontRole	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	22
EditorFonts	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	84
EditorFonts	Infrastructure	Source/Editor/Infrastructure/EditorTypography.hpp	7
EditorFontSpec	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	38
EditorFontStatus	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	67
EditorSettings	Infrastructure	Source/Editor/Infrastructure/EditorSettings.hpp	19
EditorSettingsPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	51
EditorSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	100
Entry	Infrastructure	Source/Editor/Infrastructure/TextureThumbnails.hpp	52
ExternalClientDesc	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	37
FrameComposer	RHI	Source/Engine/RHI/FrameComposer.hpp	64
FrameComposerStats	RHI	Source/Engine/RHI/FrameComposer.hpp	45
FrameGenerationMode	Infrastructure	Source/Editor/Infrastructure/	138

		ProjectConfig.hpp	
FrameGenerationPolicySetting	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	131
HudTargetMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	147
ImFont	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	10
ImGuiIO	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	11
Impl	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	221
InteractionTransaction	Infrastructure	Source/Editor/Infrastructure/EditTransactions.hpp	16
LoadedSource	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	138
MaterialGraphPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	52
MaterialGraphRuntime	Infrastructure	Source/Editor/Infrastructure/MaterialGraphRuntime.hpp	41
MaterialRuntimeStats	Infrastructure	Source/Editor/Infrastructure/MaterialGraphRuntime.hpp	25
MotionHistory2D	RHI	Source/Engine/RHI/FrameComposer.hpp	30
NetSession	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	32
NetworkDiagnosticsPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	53
OitMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	98
OpaqueRenderMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	91
PendingEdit	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	255
PendingSpawn	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	264
PhysicsDebugFlags	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	19
PhysicsDebugFrame	Infrastructure	Source/Editor/Infrastructure/	47

		PhysicsDebugDraw.hpp	
PhysicsDebugLine	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	32
PhysicsDebugPoint	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	40
PhysicsDebugVisualizer	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	58
PiePlayerInput	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	27
PlaybackIcon	Infrastructure	Source/Editor/Infrastructure/EditorWidgets.hpp	12
PlayInEditor	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	44
Profiler	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	27
ProjectConfig	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	57
ProjectConfig	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	254
ProjectConfigStore	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	54
ProjectConfigStore	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	314
ProjectDashboard	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	55
ProjectPaths	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	58
ProjectPaths	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	276
ProjectPaths	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	21
ProjectSettings	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	162
ProjectSettings	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	22
ProjectSettingsPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	56
Range	Infrastructure	Source/Editor/	249

		Infrastructure/ CommandStack.hpp	
ReleaseCookConfig	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	70
ReleaseCooker	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	154
ReleaseCookProgress	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	117
ReleaseCookResult	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	138
ReleaseCookStage	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	28
ResolvedHudSettings	Infrastructure	Source/Editor/ Infrastructure/ ResolvedRenderSettings.hpp	35
ResolvedRenderSettings	Infrastructure	Source/Editor/ Infrastructure/ ResolvedRenderSettings.hpp	14
Result	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	43
Result	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	47
SdfCollisionSystem	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	13
SdfField	Physics	Source/Engine/ Physics/ SdfCollision.hpp	20
SdfGiVolume	Physics	Source/Engine/ Physics/SdfGi.hpp	21
Services	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	220
SettingDesc	Infrastructure	Source/Editor/ Infrastructure/ SettingsSchema.hpp	28
SettingKind	Infrastructure	Source/Editor/ Infrastructure/ SettingsSchema.hpp	20
SettingScope	Infrastructure	Source/Editor/ Infrastructure/ SettingsSchema.hpp	11
Slot	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime.hpp	118
Source	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	19

SpawnComponent	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	278
SpawnHeader	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	271
SpawnTransaction	Infrastructure	Source/Editor/Infrastructure/EditTransactions.hpp	190
StableSource	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	145
Stats	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	88
Stats	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	55
Stats	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	95
TemporalUpscaleMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	154
TextureArrayAsset	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	23
TextureBinding	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	142
TextureThumbnailCache	Infrastructure	Source/Editor/Infrastructure/TextureThumbnails.hpp	17
UpscaleMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	105
UpscalePolicy	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	114
UpscalerBackendSetting	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	121
UserSettings	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	59
UserSettings	Infrastructure	Source/Editor/Infrastructure/UserSettings.hpp	16
VirtualFileSystem	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	20
VirtualFileSystem	Infrastructure	Source/Editor/Infrastructure/	15

		EditorFonts.hpp	
VirtualFileSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	18
VirtualTextureSettings	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	37
VirtualTextureSettings	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	31
VirtualTextureSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	22
VisualScriptPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	60
WindowManager	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	16

Editor/Panels (79)

Символ	Вид	Заголовок	Строка
Action	Panels	Source/Editor/Panels/ProjectDashboard.hpp	25
AIStateTreeCanvasPanel	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	55
AIStateTreeCompiled	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	35
AIStateTreeDocument	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	24
AnimationGraphCanvasPanel	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	116
AnimationGraphDocument	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	79
AnimationGraphLink	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	65
AnimationGraphLinkKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	33
AnimationGraphNode	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	45

AnimationGraphNodeKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	26
AnimationTimelinePanel	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	34
AssetImportService	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	24
AudioSystem	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	15
AudioSystem	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	11
CanvasModelOps	Panels	Source/Editor/Panels/NodeCanvasPanel.hpp	21
CommandStack	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	25
CommandStack	Panels	Source/Editor/Panels/SettingsPanels.hpp	30
ComponentRegistry	Panels	Source/Editor/Panels/ScriptDebuggerPanel.hpp	18
Config	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	35
ContentBrowserPanel	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	27
DragPayload	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	60
EditorFonts	Panels	Source/Editor/Panels/ProjectDashboard.hpp	20
EditorFonts	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	20
EditorSettingsPanel	Panels	Source/Editor/Panels/SettingsPanels.hpp	91
EditSource	Panels	Source/Editor/Panels/AIStateTreeCanvasPa	58

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EditSource	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	119
Entry	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	43
FileDialog	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	140
ImNodesContext	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	13
ImNodesContext	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	15
LevelSettingsPanel	Panels	Source/Editor/Panels/LevelSettingsPanel.hpp	12
LicensesPanel	Panels	Source/Editor/Panels/LicensesPanel.hpp	11
LogBoyPanel	Panels	Source/Editor/Panels/LogBoyPanel.hpp	394
LogBoyRecord	Panels	Source/Editor/Panels/LogBoyPanel.hpp	105
LogBoyRing	Panels	Source/Editor/Panels/LogBoyPanel.hpp	121
LogBoySeverity	Panels	Source/Editor/Panels/LogBoyPanel.hpp	33
LogBoySystem	Panels	Source/Editor/Panels/LogBoyPanel.hpp	186
LogicGraphDocument	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	23
MaterialGraphPanel	Panels	Source/Editor/Panels/MaterialGraphPanel.hpp	20
Mode	Panels	Source/Editor/Panels/ProjectDashboard.hpp	84
NetSession	Panels	Source/Editor/Panels/NetworkDiagnosticsPanel.hpp	6
NetworkDiagnosticsP	Panels	Source/Editor/	15

anel		Panels/ NetworkDiagnosticsPanel.hpp	
NodeCanvasPanel	Panels	Source/Editor/ Panels/ NodeCanvasPanel.hpp	54
PIEAnimFrame	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	101
PieClientViewport	Panels	Source/Editor/ Panels/ PieClientViewport.hpp	15
PIEFrameSlot	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	289
ProjectConfigStore	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	31
ProjectConfigStore	Panels	Source/Editor/ Panels/ VoiceDevicePanel.hpp	17
ProjectDashboard	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	22
ProjectSettingsPanel	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	33
QueryKind	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	430
SceneDiagnosticsPanel	Panels	Source/Editor/ Panels/ SceneDiagnosticsPanel.hpp	56
SceneDiagnosticsSample	Panels	Source/Editor/ Panels/ SceneDiagnosticsPanel.hpp	15
ScriptDebuggerPanel	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel.hpp	28
ScriptRuntime	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	16
SinkUser	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	88
SkeletonInspectorState	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.hpp	19
SpawnContext	Panels	Source/Editor/	68

		Panels/ ContentBrowserPanel .hpp	
SpriteAnimationInsp ectorState	Panels	Source/Editor/ Panels/ SpriteAnimationPane l.hpp	21
SystemContext	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	24
SystemContext	Panels	Source/Editor/ Panels/ SkeletonAssetPanel. hpp	13
SystemContext	Panels	Source/Editor/ Panels/ SpriteAnimationPane l.hpp	13
SystemName	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	189
TextPreviewPanel	Panels	Source/Editor/ Panels/ TextPreviewPanel.hp p	24
TextureThumbnailCac he	Panels	Source/Editor/ Panels/ SpriteAnimationPane l.hpp	19
TilemapEditorPanel	Panels	Source/Editor/ Panels/ TilemapEditorPanel. hpp	20
TimelinePanelContex t	Panels	Source/Editor/ Panels/ AnimationTimelinePa nel.hpp	22
TransportGroupPanel	Panels	Source/Editor/ Panels/ TransportGroupPanel .hpp	14
UserSettingsPanel	Panels	Source/Editor/ Panels/ UserSettingsPanel.h pp	16
VirtualFileSystem	Panels	Source/Editor/ Panels/ ContentBrowserPanel .hpp	18
VirtualFileSystem	Panels	Source/Editor/ Panels/ ProjectDashboard.hp p	13
VirtualFileSystem	Panels	Source/Editor/ Panels/ TextPreviewPanel.hp p	15
VirtualTexturePanel	Panels	Source/Editor/ Panels/ VirtualTexturePanel	18

		.hpp	
VirtualTextureSystem	Panels	Source/Editor/Panels/VirtualTexturePanel.hpp	10
VisualScriptPanel	Panels	Source/Editor/Panels/ScriptDebuggerPanel.hpp	24
VisualScriptPanel	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	32
VoiceDevicePanel	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	19
WidgetCanvas	Panels	Source/Editor/Panels/SettingsPanels.hpp	18
WindowManager	Panels	Source/Editor/Panels/ProjectDashboard.hpp	14

Engine/AI (6)

Символ	Вид	Заголовок	Строка
AIAgentSettings	AI	Source/Engine/AI/AIController.hpp	16
AIControllerComponent	AI	Source/Engine/AI/AIController.hpp	31
AIControllerSystem	AI	Source/Engine/AI/AIController.hpp	84
AIPerceptionComponent	AI	Source/Engine/AI/AIController.hpp	59
AIPerceptionSettings	AI	Source/Engine/AI/AIController.hpp	48
AIPerceptionSystem	AI	Source/Engine/AI/AIController.hpp	67

Engine/Assets (58)

Символ	Вид	Заголовок	Строка
AeadAlgorithm	Assets	Source/Engine/Assets/FemCrypto.hpp	18
AeadKey	Assets	Source/Engine/Assets/FemCrypto.hpp	39
AeadNonce	Assets	Source/Engine/Assets/FemCrypto.hpp	46
AeadTag	Assets	Source/Engine/Assets/FemCrypto.hpp	54
ArchiveBuildOptions	Assets	Source/Engine/Assets/	99

		FemArchiver.hpp	
ArchiveCipher	Assets	Source/Engine/ Assets/ FemArchiver.hpp	29
ArchiveCodec	Assets	Source/Engine/ Assets/ FemArchiver.hpp	22
ArchiveEntryView	Assets	Source/Engine/ Assets/ FemArchiver.hpp	115
ArchiveInput	Assets	Source/Engine/ Assets/ FemArchiver.hpp	72
ArchiveProgress	Assets	Source/Engine/ Assets/ FemArchiver.hpp	85
ArchiveProgressStage	Assets	Source/Engine/ Assets/ FemArchiver.hpp	78
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	126
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	145
ArchiveStatus	Assets	Source/Engine/ Assets/ FemArchiver.hpp	36
AssetDatabase	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	28
AssetDependency	Assets	Source/Engine/ Assets/ AssetMeta.hpp	83
AssetHandle	Assets	Source/Engine/ Assets/ AssetMeta.hpp	22
AssetImportSettings	Assets	Source/Engine/ Assets/ AssetMeta.hpp	45
AssetMeta	Assets	Source/Engine/ Assets/ AssetMeta.hpp	61
AssetRecord	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	19
AstcBlock	Assets	Source/Engine/ Assets/ TextureCook.hpp	25
AtlasDesc	Assets	Source/Engine/ Assets/ TextureArray.hpp	31
AtlasLayer	Assets	Source/Engine/ Assets/ TextureArray.hpp	21
AudioClipHeader	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	87
BlockExtent	Assets	Source/Engine/ Assets/	103

		TextureFormat.hpp	
CleanRenderExtract	Assets	Source/Engine/ Assets/ ProjectScaffold.hpp	127
CookedAudioView	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	116
CookedTexture	Assets	Source/Engine/ Assets/ TextureCook.hpp	41
CookedTextureView	Assets	Source/Engine/ Assets/ TextureFormat.hpp	203
CookProgressFlags	Assets	Source/Engine/ Assets/ CookProgress.hpp	31
CookProgressRecord	Assets	Source/Engine/ Assets/ CookProgress.hpp	38
CookProgressStage	Assets	Source/Engine/ Assets/ CookProgress.hpp	18
CookTarget	Assets	Source/Engine/ Assets/ TextureCook.hpp	17
FanmHeader	Assets	Source/Engine/ Assets/ AnimationClipAsset. hpp	22
FemArchiver	Assets	Source/Engine/ Assets/ FemArchiver.hpp	128
FempakInput	Assets	Source/Engine/ Assets/ FempakWriter.hpp	15
FempakWriter	Assets	Source/Engine/ Assets/ FempakWriter.hpp	22
LocaleTable	Assets	Source/Engine/ Assets/ Localization.hpp	14
LocalizationGender	Assets	Source/Engine/ Assets/ Localization.hpp	21
LocalizationPack	Assets	Source/Engine/ Assets/ Localization.hpp	13
LocalizationTable	Assets	Source/Engine/ Assets/ Localization.hpp	28
MappedFile	Assets	Source/Engine/ Assets/VFS.hpp	19
MountRecord	Assets	Source/Engine/ Assets/VFS.hpp	128
PakEntry	Assets	Source/Engine/ Assets/ PakFormat.hpp	26
PakHeader	Assets	Source/Engine/ Assets/ PakFormat.hpp	16

PcmFormat	Assets	Source/Engine/Assets/AudioClipFormat.hpp	21
PinnedFile	Assets	Source/Engine/Assets/VFS.hpp	63
ReloadNotice	Assets	Source/Engine/Assets/AssetDatabase.hpp	86
StagedRow	Assets	Source/Engine/Assets/AssetDatabase.hpp	108
StringEntry	Assets	Source/Engine/Assets/Localization.hpp	15
TextureAlphaMode	Assets	Source/Engine/Assets/TextureFormat.hpp	43
TextureArrayAsset	Assets	Source/Engine/Assets/TextureArray.hpp	51
TextureCookOptions	Assets	Source/Engine/Assets/TextureCook.hpp	31
TextureFormat	Assets	Source/Engine/Assets/TextureFormat.hpp	22
TextureHeader	Assets	Source/Engine/Assets/TextureFormat.hpp	184
VfsEntry	Assets	Source/Engine/Assets/VFS.hpp	55
VirtualFileSystem	Assets	Source/Engine/Assets/TextureArray.hpp	16
VirtualFileSystem	Assets	Source/Engine/Assets/VFS.hpp	69

Engine/Audio (101)

Символ	Вид	Заголовок	Строка
AudioBus	Audio	Source/Engine/Audio/Audio.hpp	26
AudioBusComponent	Audio	Source/Engine/Audio/Audio.hpp	40
AudioCaptureMonitor	Audio	Source/Engine/Audio/AudioDevices.hpp	144
AudioClip	Audio	Source/Engine/Audio/Audio.hpp	69
AudioCommand	Audio	Source/Engine/Audio/DspGraph.hpp	138
AudioCommandType	Audio	Source/Engine/Audio/DspGraph.hpp	121
AudioContextComponent	Audio	Source/Engine/Audio/AudioZones.hpp	67
AudioDeviceEnumerator	Audio	Source/Engine/Audio/AudioDevices.hpp	93

AudioDeviceInfo	Audio	Source/Engine/ Audio/ AudioDevices.hpp	54
AudioDeviceKind	Audio	Source/Engine/ Audio/ AudioDevices.hpp	15
AudioDeviceList	Audio	Source/Engine/ Audio/ AudioDevices.hpp	66
AudioDevicePreferences	Audio	Source/Engine/ Audio/ AudioDevices.hpp	73
AudioDeviceStatus	Audio	Source/Engine/ Audio/ AudioDevices.hpp	21
AudioLimiter	Audio	Source/Engine/ Audio/Limiter.hpp	23
AudioListener	Audio	Source/Engine/ Audio/Audio.hpp	94
AudioSource	Audio	Source/Engine/ Audio/Audio.hpp	79
AudioSourceComponent	Audio	Source/Engine/ Audio/Audio.hpp	53
AudioSystem	Audio	Source/Engine/ Audio/Audio.hpp	100
AudioZoneComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	32
AudioZoneKind	Audio	Source/Engine/ Audio/ AudioZones.hpp	14
AudioZoneSystem	Audio	Source/Engine/ Audio/ AudioZones.hpp	91
BiquadState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	369
BusAtomics	Audio	Source/Engine/ Audio/DspGraph.hpp	370
BusFilterParams	Audio	Source/Engine/ Audio/DspGraph.hpp	108
BusRuntime	Audio	Source/Engine/ Audio/Audio.hpp	212
BusShadow	Audio	Source/Engine/ Audio/DspGraph.hpp	291
BusState	Audio	Source/Engine/ Audio/DspGraph.hpp	338
CommandQueue	Audio	Source/Engine/ Audio/DspGraph.hpp	170
CookedClipRecord	Audio	Source/Engine/ Audio/Audio.hpp	187
DspClipView	Audio	Source/Engine/ Audio/DspGraph.hpp	205
DspGraph	Audio	Source/Engine/ Audio/DspGraph.hpp	216
DspGraph	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	16
DspResult	Audio	Source/Engine/ Audio/DspGraph.hpp	34

DspResult	Audio	Source/Engine/Audio/VoiceChannel.hpp	19
DuckingParams	Audio	Source/Engine/Audio/DspGraph.hpp	80
DuckingState	Audio	Source/Engine/Audio/DspGraph.hpp	298
FilterState	Audio	Source/Engine/Audio/DspGraph.hpp	327
FloatRing	Audio	Source/Engine/Audio/Audio.hpp	24
FloatRing	Audio	Source/Engine/Audio/AudioStream.hpp	16
FloatRing	Audio	Source/Engine/Audio/DspGraph.hpp	32
JitterSlot	Audio	Source/Engine/Audio/VoiceChannel.hpp	119
LimiterDesc	Audio	Source/Engine/Audio/Limiter.hpp	11
OggOpusStream	Audio	Source/Engine/Audio/OggOpus.hpp	131
OggPacket	Audio	Source/Engine/Audio/OggOpus.hpp	76
OggPacketReader	Audio	Source/Engine/Audio/OggOpus.hpp	86
OggPage	Audio	Source/Engine/Audio/OggOpus.hpp	36
OggPageReader	Audio	Source/Engine/Audio/OggOpus.hpp	57
OpusHeadInfo	Audio	Source/Engine/Audio/OggOpus.hpp	106
OwnedPcmRecord	Audio	Source/Engine/Audio/Audio.hpp	194
Peer	Audio	Source/Engine/Audio/VoiceChannel.hpp	129
RateBucket	Audio	Source/Engine/Audio/VoiceNetworkBridge.hpp	62
ResolvedZone	Audio	Source/Engine/Audio/AudioZones.hpp	113
ReverbParams	Audio	Source/Engine/Audio/DspGraph.hpp	96
ReverbState	Audio	Source/Engine/Audio/DspGraph.hpp	311
SlotRuntime	Audio	Source/Engine/Audio/VoiceEffects.hpp	385
SourceRuntime	Audio	Source/Engine/Audio/Audio.hpp	204
State	Audio	Source/Engine/Audio/AudioDevices.hpp	164
Stats	Audio	Source/Engine/Audio/AudioZones.hpp	94

VoiceAec	Audio	Source/Engine/ Audio/VoiceAec.hpp	44
VoiceAecSettings	Audio	Source/Engine/ Audio/VoiceAec.hpp	22
VoiceCapture	Audio	Source/Engine/ Audio/ AudioStream.hpp	64
VoiceChannel	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	65
VoiceChannelConfig	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	36
VoiceChannelStats	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	46
VoiceCodecConfig	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	15
VoiceConditionOutput	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	333
VoiceConditionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	284
VoiceConditionState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	324
VoiceDecoder	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	51
VoiceDenoiseTier	Audio	Source/Engine/ Audio/VoiceDsp.hpp	10
VoiceDistortionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	157
VoiceDspChain	Audio	Source/Engine/ Audio/VoiceDsp.hpp	99
VoiceDspSettings	Audio	Source/Engine/ Audio/VoiceDsp.hpp	43
VoiceEffectChain	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	345
VoiceEffectChainSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	221
VoiceEffectKind	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	20
VoiceEffectPresetKind	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	235
VoiceEffectProfile	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	310
VoiceEffectRack	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	433
VoiceEffectSlot	Audio	Source/Engine/ Audio/	200

		VoiceEffects.hpp	
VoiceEncoder	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	27
VoiceFormantSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	146
VoiceMuffleSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	80
VoiceMuteSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	67
VoiceNetworkBridge	Audio	Source/Engine/ Audio/ VoiceNetworkBridge.hpp	18
VoiceParticipant	Audio	Source/Engine/ Audio/ VoiceSession.hpp	46
VoicePitchSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	133
VoicePresetKind	Audio	Source/Engine/ Audio/VoiceDsp.hpp	68
VoiceRadioSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	115
VoiceRingModulationSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	171
VoiceRouting	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	30
VoiceSession	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	15
VoiceSession	Audio	Source/Engine/ Audio/ VoiceSession.hpp	65
VoiceSpatialPosition	Audio	Source/Engine/ Audio/ VoiceSession.hpp	37
VoiceSpatialSettings	Audio	Source/Engine/ Audio/ VoiceSession.hpp	22
VoiceState	Audio	Source/Engine/ Audio/DspGraph.hpp	350
VoiceSystem	Audio	Source/Engine/ Audio/ VoiceSystem.hpp	39
VoiceSystemConfig	Audio	Source/Engine/ Audio/ VoiceSystem.hpp	22
VoiceUnderwaterSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	98
VoiceWhisperSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	185

VorbisIdentInfo	Audio	Source/Engine/Audio/OggOpus.hpp	121

Engine/Core (103)

Символ	Вид	Заголовок	Строка
AssetId	Core	Source/Engine/Core/AssetId.hpp	43
AssetKind	Core	Source/Engine/Core/AssetId.hpp	15
BigIntT	Core	Source/Engine/Core/BigInt.hpp	179
BigNumberStore	Core	Source/Engine/Core/BigNumberStore.hpp	41
BinaryKind	Core	Source/Engine/Core/BigNumberStore.hpp	259
BinaryKind	Core	Source/Engine/Core/DecimalStore.hpp	206
BufferTag	Core	Source/Engine/Core/Types.hpp	301
BuildInfo	Core	Source/Engine/Core/BuildInfo.hpp	12
BuildProfile	Core	Source/Engine/Core/BuildProfile.hpp	135
CameraView	Core	Source/Engine/Core/Grid.hpp	337
Cell	Core	Source/Engine/Core/Grid.hpp	74
CivilDate	Core	Source/Engine/Core/Calendar.hpp	21
Clock	Core	Source/Engine/Core/Time.hpp	28
ClockTime	Core	Source/Engine/Core/Calendar.hpp	32
Color32	Core	Source/Engine/Core/Types.hpp	203
ComponentDesc	Core	Source/Engine/Core/Mirror.hpp	48
ComponentRegistry	Core	Source/Engine/Core/PluginHost.hpp	27
ComponentType	Core	Source/Engine/Core/PluginHost.hpp	18
DateTime	Core	Source/Engine/Core/Calendar.hpp	44
DecimalStore	Core	Source/Engine/Core/DecimalStore.hpp	17
DecimalT	Core	Source/Engine/Core/Decimal.hpp	34
DurationStyle	Core	Source/Engine/Core/Time.hpp	571
EddsaKeyPair	Core	Source/Engine/Core/CryptoAsymmetric.hpp	31

FieldDesc	Core	Source/Engine/Core/Mirror.hpp	37
FieldKind	Core	Source/Engine/Core/Mirror.hpp	17
FileFilter	Core	Source/Engine/Core/WindowManager.hpp	183
FormatSpec	Core	Source/Engine/Core/BigInt.hpp	36
FrameArena	Core	Source/Engine/Core/FrameArena.hpp	12
FrameTimer	Core	Source/Engine/Core/Time.hpp	501
GamepadState	Core	Source/Engine/Core/WindowManager.hpp	26
GridDesc	Core	Source/Engine/Core/Grid.hpp	91
Handle	Core	Source/Engine/Core/BigNumberStore.hpp	20
Handle	Core	Source/Engine/Core/Types.hpp	275
Handle	Core	Source/Engine/Core/WindowHost.hpp	20
Host	Core	Source/Engine/Core/Plugin.hpp	20
InputState	Core	Source/Engine/Core/WindowManager.hpp	114
JobSystem	Core	Source/Engine/Core/JobSystem.hpp	25
Key	Core	Source/Engine/Core/WindowManager.hpp	46
KeyLess	Core	Source/Engine/Core/FlatMap.hpp	203
LoadedPlugin	Core	Source/Engine/Core/PluginHost.hpp	64
LogLevel	Core	Source/Engine/Core/Log.hpp	13
LogRetention	Core	Source/Engine/Core/Log.hpp	31
Mat4	Core	Source/Engine/Core/Mat4.hpp	14
NoiseGenerator	Core	Source/Engine/Core/NoiseGenerator.hpp	10
Notation	Core	Source/Engine/Core/BigInt.hpp	30
NoticeEntry	Core	Source/Engine/Core/ThirdPartyNotices.hpp	12
OptimizationAdvice	Core	Source/Engine/Core/BuildProfile.hpp	154
OptimizationLevel	Core	Source/Engine/Core/BuildProfile.hpp	10

Pcg32	Core	Source/Engine/Core/Pcg32.hpp	20
PipelineTag	Core	Source/Engine/Core/Types.hpp	303
PluginHost	Core	Source/Engine/Core/PluginHost.hpp	76
PluginSystem	Core	Source/Engine/Core/PluginHost.hpp	54
Profiler	Core	Source/Engine/Core/Profiler.hpp	43
ProfileRow	Core	Source/Engine/Core/ProfilerPanel.hpp	26
ProfilerPanel	Core	Source/Engine/Core/ProfilerPanel.hpp	39
ProfileSample	Core	Source/Engine/Core/Profiler.hpp	28
ProfileSortMode	Core	Source/Engine/Core/ProfilerPanel.hpp	16
ProfileStats	Core	Source/Engine/Core/Profiler.hpp	16
ProjectionMode	Core	Source/Engine/Core/Grid.hpp	17
Rect	Core	Source/Engine/Core/Types.hpp	173
Reflection	Core	Source/Engine/Core/Mirror.hpp	66
Reflection	Core	Source/Engine/Core/Mirror.hpp	294
Reflection	Core	Source/Engine/Core/Mirror.hpp	306
ReplayFrame	Core	Source/Engine/Core/Replay.hpp	238
ReplayKeyframe	Core	Source/Engine/Core/Replay.hpp	16
ReplaySession	Core	Source/Engine/Core/Replay.hpp	25
RngState	Core	Source/Engine/Core/Pcg32.hpp	11
RoundingMode	Core	Source/Engine/Core/Decimal.hpp	22
SamplerTag	Core	Source/Engine/Core/Types.hpp	304
Scope	Core	Source/Engine/Core/Profiler.hpp	141
SecureRng	Core	Source/Engine/Core/SecureRng.hpp	11
Sha256Context	Core	Source/Engine/Core/Sha256.hpp	17
Slot	Core	Source/Engine/Core/BigNumberStore.hpp	252
Slot	Core	Source/Engine/Core/DecimalStore.hpp	199
SplashClock	Core	Source/Engine/Core/Splash.hpp	10
Status	Core	Source/Engine/	18

		Core/BigInt.hpp	
Stopwatch	Core	Source/Engine/ Core/Time.hpp	209
Task	Core	Source/Engine/ Core/JobSystem.hpp	96
TextureTag	Core	Source/Engine/ Core/Types.hpp	302
TextWriter	Core	Source/Engine/ Core/Calendar.hpp	189
ThreadCounter	Core	Source/Engine/ Core/JobSystem.hpp	109
ThreadLoad	Core	Source/Engine/ Core/JobSystem.hpp	31
TimerFired	Core	Source/Engine/ Core/Time.hpp	255
TimerHandle	Core	Source/Engine/ Core/Time.hpp	240
TimerPool	Core	Source/Engine/ Core/Time.hpp	266
TypeNameOverride	Core	Source/Engine/ Core/ Definitions.hpp	40
TypeNameOverride	Core	Source/Engine/ Core/ Definitions.hpp	283
TypeObject	Core	Source/Engine/ Core/ Definitions.hpp	185
TypeRecord	Core	Source/Engine/ Core/ Definitions.hpp	85
TypeTable	Core	Source/Engine/ Core/ Definitions.hpp	93
TypeTraits	Core	Source/Engine/ Core/ Definitions.hpp	170
TypeTraits	Core	Source/Engine/ Core/ Definitions.hpp	267
Unregistered	Core	Source/Engine/ Core/Mirror.hpp	59
Vector2D	Core	Source/Engine/ Core/Types.hpp	13
Vector3D	Core	Source/Engine/ Core/Types.hpp	90
Vector4D	Core	Source/Engine/ Core/Types.hpp	139
WindowDesc	Core	Source/Engine/ Core/ WindowManager.hpp	204
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GameplayAttributeRegistry	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	35
GameplayAttributesComponent	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	66
GameplayAttributeValues	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	60
GameplayEffectDef	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	45
GameplayEffectEvent	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	95
GameplayEffectEventType	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	88
GameplayEffectFlags	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	32
GameplayEffectsComponent	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	78
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GameplayEffectSystem	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	107
GameplayModifierOp	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	75
GameplaySkillDef	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	55
GameplaySkillEvent	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	98
GameplaySkillEventType	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	33
GameplaySkillFailReason	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	41
GameplaySkillsComponent	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	89
GameplaySkillSlot	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	77
GameplaySkillState	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	27
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Engine/Input (14)

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InputContextDesc	Input	Source/Engine/Input/InputSystem.hpp	87
InputEvent	Input	Source/Engine/Input/InputSystem.hpp	120
InputEventType	Input	Source/Engine/Input/InputSystem.hpp	60
InputMappingDesc	Input	Source/Engine/Input/InputSystem.hpp	94
InputModifier	Input	Source/Engine/Input/InputSystem.hpp	74
InputModifierKind	Input	Source/Engine/Input/InputSystem.hpp	39
InputSource	Input	Source/Engine/Input/InputSystem.hpp	68
InputSourceKind	Input	Source/Engine/Input/	30

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InputTriggerKind	Input	Source/Engine/ Input/ InputSystem.hpp	50
UserSlot	Input	Source/Engine/ Input/ InputSystem.hpp	175

Engine/Navigation (25)

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CrowdPool	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	46
CrowdRenderSettings	Navigation	Source/Engine/ Navigation/ CrowdRender.hpp	14
CrowdSettings	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	27
Entry	Navigation	Source/Engine/ Navigation/ FlowField.hpp	79
FlowField	Navigation	Source/Engine/ Navigation/ FlowField.hpp	33
FlowFieldDesc	Navigation	Source/Engine/ Navigation/ FlowField.hpp	22
NavAgentComponent	Navigation	Source/Engine/ Navigation/ Navigation.hpp	242
NavAgentPathSettings	Navigation	Source/Engine/ Navigation/ Navigation.hpp	206
NavAgentSettings	Navigation	Source/Engine/ Navigation/ Navigation.hpp	184
NavigationSystem	Navigation	Source/Engine/ Navigation/ Navigation.hpp	265
NavMap	Navigation	Source/Engine/ Navigation/ Navigation.hpp	27
NavMesh	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	49
NavMesh	Navigation	Source/Engine/ Navigation/ Navigation.hpp	18
NavMeshPortal	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	39

NavMeshRect	Navigation	Source/Engine/Navigation/NavMesh.hpp	19
NavPath	Navigation	Source/Engine/Navigation/Navigation.hpp	99
NavPathOptions	Navigation	Source/Engine/Navigation/Navigation.hpp	80
NavSmoothingMode	Navigation	Source/Engine/Navigation/Navigation.hpp	72
Page	Navigation	Source/Engine/Navigation/PagedNavGrid.hpp	105
PagedNavGrid	Navigation	Source/Engine/Navigation/PagedNavGrid.hpp	32
PagedNavGridDesc	Navigation	Source/Engine/Navigation/PagedNavGrid.hpp	23
PathFinder	Navigation	Source/Engine/Navigation/Navigation.hpp	127
PathStats	Navigation	Source/Engine/Navigation/Navigation.hpp	116
TilemapComponent	Navigation	Source/Engine/Navigation/PagedNavGrid.hpp	14

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Endpoint	Network	Source/Engine/Network/NetLink.hpp	92
Frame	Network	Source/Engine/Network/NetLink.hpp	84
FrameInfo	Network	Source/Engine/Network/NetRollback.hpp	221
FrameMeta	Network	Source/Engine/Network/NetRollback.hpp	473
InputChannel	Network	Source/Engine/Network/NetProtocol.hpp	56

InputChannelStats	Network	Source/Engine/Network/NetProtocol.hpp	44
InputPacket	Network	Source/Engine/Network/NetProtocol.hpp	24
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LobbyError	Network	Source/Engine/Network/Lobby.hpp	37
LobbyMessageType	Network	Source/Engine/Network/Lobby.hpp	25
LobbyRoomInfo	Network	Source/Engine/Network/Lobby.hpp	73
LobbyServer	Network	Source/Engine/Network/Lobby.hpp	93
LobbyServerConfig	Network	Source/Engine/Network/Lobby.hpp	82
LocalEntry	Network	Source/Engine/Network/NetRollback.hpp	205
Member	Network	Source/Engine/Network/Lobby.hpp	126
MemoryEndpoint	Network	Source/Engine/Network/NetLink.hpp	46
MemoryEndpoint	Network	Source/Engine/Network/NetLink.hpp	113
MemoryNetwork	Network	Source/Engine/Network/NetLink.hpp	53
NetAddress	Network	Source/Engine/Network/Transport.hpp	19
NetDiagnosticsSnapshot	Network	Source/Engine/Network/Diagnostics.hpp	12
NetDisconnectReason	Network	Source/Engine/Network/Session.hpp	65
NetEntityId	Network	Source/Engine/Network/NetRollback.hpp	39
NetEntityMap	Network	Source/Engine/Network/NetRollback.hpp	167
NetEntry	Network	Source/Engine/Network/NetRollback.hpp	200
NetIdentity	Network	Source/Engine/Network/Identity.hpp	32
NetInputFrame	Network	Source/Engine/Network/NetRollback.hpp	70
NetInputStatus	Network	Source/Engine/Network/NetRollback.hpp	99
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NetLobbyClientConfig	Network	Source/Engine/Network/Lobby.hpp	151
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NetPlayerSlot	Network	Source/Engine/Network/Session.hpp	134
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NetSession	Network	Source/Engine/Network/Session.hpp	189
NetSessionCallbacks	Network	Source/Engine/Network/Session.hpp	178
NetSessionConfig	Network	Source/Engine/Network/Lobby.hpp	15
NetSessionConfig	Network	Source/Engine/Network/Session.hpp	149
NetSessionMode	Network	Source/Engine/Network/Session.hpp	22
NetSessionStatus	Network	Source/Engine/Network/Session.hpp	46
NetSlotState	Network	Source/Engine/Network/Session.hpp	58
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NetTickMode	Network	Source/Engine/Network/Session.hpp	40
NetTransport	Network	Source/Engine/Network/Transport.hpp	127
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Peer	Network	Source/Engine/Network/SecureSessionTransport.hpp	87
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RelevanceConfig	Network	Source/Engine/ Network/ RelevanceGrid.hpp	56
RelevanceEntity	Network	Source/Engine/ Network/ RelevanceGrid.hpp	81
RelevanceGrid	Network	Source/Engine/ Network/ NetRollback.hpp	36
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RelevancePlan	Network	Source/Engine/ Network/ RelevanceGrid.hpp	93
ReplicationStatus	Network	Source/Engine/ Network/ Replication.hpp	28
Result	Network	Source/Engine/ Network/ NetRollback.hpp	243
RollbackReport	Network	Source/Engine/ Network/ NetRollback.hpp	229
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FormatEntry	Text	Source/Engine/ Text/ TextService.hpp	583
FormattedLayoutEntry	Text	Source/Engine/ Text/ TextService.hpp	566
FormattedLayoutKey	Text	Source/Engine/ Text/ TextService.hpp	556
HarfBuzzShaper	Text	Source/Engine/ Text/TextShaper.hpp	110
Impl	Text	Source/Engine/ Text/TextShaper.hpp	128
ITextShaper	Text	Source/Engine/ Text/TextShaper.hpp	89
LayoutEntry	Text	Source/Engine/ Text/ TextService.hpp	548
LayoutKey	Text	Source/Engine/ Text/ TextService.hpp	528
MappedTextAsset	Text	Source/Engine/ Text/ TextAssetFile.hpp	17
MarkupStackEntry	Text	Source/Engine/ Text/TextMarkup.hpp	71
MarkupVariantEntry	Text	Source/Engine/ Text/ TextService.hpp	574
ScriptBackend	Text	Source/Engine/ Text/ TextShapeSystem.hpp	111
ScriptShaper	Text	Source/Engine/ Text/TextShaper.hpp	207
ShapedGlyph	Text	Source/Engine/ Text/TextShaper.hpp	56
ShapedGlyphHit	Text	Source/Engine/ Text/	153

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ShapedLayoutResult	Text	Source/Engine/Text/TextShapedLayout.hpp	26
ShapedLine	Text	Source/Engine/Text/TextShapedLayout.hpp	18
StyledAsset	Text	Source/Engine/Text/TextService.hpp	595
TextAction	Text	Source/Engine/Text/TextStyle.hpp	61
TextActionKind	Text	Source/Engine/Text/TextStyle.hpp	51
TextAnimationParams	Text	Source/Engine/Text/TextRender.hpp	29
TextAssetBlock	Text	Source/Engine/Text/TextAssetFormat.hpp	34
TextAssetHeader	Text	Source/Engine/Text/TextAssetFormat.hpp	20
TextAssetRef	Text	Source/Engine/Text/TextAsset.hpp	14
TextAssetStatus	Text	Source/Engine/Text/TextAssetFormat.hpp	63
TextAssetView	Text	Source/Engine/Text/TextAssetFormat.hpp	42
TextBlock	Text	Source/Engine/Text/TextAsset.hpp	37
TextDirection	Text	Source/Engine/Text/TextShaper.hpp	21
TextFontStack	Text	Source/Engine/Text/TextFontStack.hpp	54
TextFormatArg	Text	Source/Engine/Text/TextFormat.hpp	45
TextFormatStatus	Text	Source/Engine/Text/TextFormat.hpp	33
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TextLayoutParams	Text	Source/Engine/Text/TextLayout.hpp	54
TextLayoutResult	Text	Source/Engine/Text/TextLayout.hpp	72
TextLine	Text	Source/Engine/Text/TextLayout.hpp	22

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TextRun	Text	Source/Engine/Text/TextShaper.hpp	47
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TextScript	Text	Source/Engine/Text/TextShaper.hpp	30
TextService	Text	Source/Engine/Text/TextService.hpp	18
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TextShapeHandle	Text	Source/Engine/Text/TextShapeCache.hpp	19
TextShapeKey	Text	Source/Engine/Text/TextShapeCache.hpp	32
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TextStore	Text	Source/Engine/Text/TextStore.hpp	19
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AccessibilityOverlay	UI	Source/Engine/UI/AccessibilityOverlay.hpp	30
AccessibilityPresentationSettings	UI	Source/Engine/UI/Accessibility.hpp	233
AccessibilityProfile	UI	Source/Engine/UI/Accessibility.hpp	272
AccessibilitySettings	UI	Source/Engine/UI/Accessibility.hpp	217
AudioSystem	UI	Source/Engine/UI/Accessibility.hpp	26
AvSpeechTts	UI	Source/Engine/UI/Tts.hpp	60
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CardBoardComponent	UI	Source/Engine/UI/CardBoard.hpp	49
CardBoardSystem	UI	Source/Engine/UI/CardBoard.hpp	74
ColorMatrix3	UI	Source/Engine/UI/Accessibility.hpp	42
ColorVisionDeficiency	UI	Source/Engine/UI/Accessibility.hpp	34
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FlashGuard	UI	Source/Engine/UI/Accessibility.hpp	59
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FontAtlas	UI	Source/Engine/UI/FontAtlas.hpp	101
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FontBuildOptions	UI	Source/Engine/UI/FontAtlas.hpp	25
FontBuildOptions	UI	Source/Engine/UI/FontBuilder.hpp	15
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FontFamilyManifest	UI	Source/Engine/UI/FontFamilyAsset.hpp	17
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FontStyle	UI	Source/Engine/UI/FontAtlas.hpp	65

FrameOutput	UI	Source/Engine/UI/PinkUI.hpp	1418
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HypertextFlags	UI	Source/Engine/UI/Hypertext.hpp	21
HypertextPool	UI	Source/Engine/UI/Hypertext.hpp	74
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LocalizationTable	UI	Source/Engine/UI/Hypertext.hpp	16
LocalizationTable	UI	Source/Engine/UI/PinkUI.hpp	39
NarrativeArchetype	UI	Source/Engine/UI/NarrativeTree.hpp	23
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NarrativeGroup	UI	Source/Engine/UI/NarrativeTree.hpp	37
NarrativeLocale	UI	Source/Engine/UI/NarrativeTree.hpp	17
NarrativeLog	UI	Source/Engine/UI/Accessibility.hpp	99
NarrativeNode	UI	Source/Engine/UI/NarrativeTree.hpp	47
NarrativeTree	UI	Source/Engine/UI/NarrativeTree.hpp	55
OledShiftState	UI	Source/Engine/UI/PinkUI.hpp	463
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PinkSdfPrimitive	UI	Source/Engine/UI/PinkSdf.hpp	33
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21. Приложение B - макросы и платформенные define

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